Integrated Analysis Report

Financial Data Analysis in Python

DCS520

Overview: In this week 6 of Financial Data Analysis in Python mini project, the chosen dataset containing historical price of cryptocurrency Dogecoin including Price Statistics (High, Low, Open, Close), Volume and Market Cap has been previously analyzed and summarized in form of Univariate Analysis, Bivariate Analysis (EDA), statistical analysis in the form of testes including T-Test, ANOVA and Linear Regression to compare the mean of features with other aspects of the dataset. Time Series Analysis and Forecasting Models such as ARIMA (Auto Regression Integrated Moving Average) has also been applied, and outcome has been analyzed.

Exploratory Data Analysis: The Dataset containing several numeric features for the price, volume and market capitalization of the cryptocurrency. Univariate and Bivariate analysis has been used to explore the data, Key findings are as follows:

- Volume and Market Cap both matrices have large ranges highlighting periods of significant market movements
- Price (High, Low, Open, Close) variables exhibit significant variability with higher valued reflecting high trading activity.

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Basic Column wise Summary of the Database such as Mean, Median, SD etc is as Follows:

                SNo         High          Low         Open        Close  \
count  2760.000000  2760.000000  2760.000000  2760.000000  2760.000000
mean   1380.500000     0.014964     0.012402     0.013669     0.013763
std     796.887696     0.069103     0.055536     0.062347     0.062559
min       1.000000     0.000089     0.000085     0.000087     0.000087
25%     690.750000     0.000231     0.000222     0.000227     0.000226
50%    1380.500000     0.001914     0.001772     0.001849     0.001844
75%    2070.250000     0.002986     0.002796     0.002900     0.002899
max    2760.000000     0.737567     0.608168     0.687801     0.684777

             Volume     Marketcap
count  2.760000e+03  2.760000e+03
mean   4.326321e+08  1.760533e+09
std    2.807320e+09  8.115186e+09
min    0.000000e+00  1.509085e+06
25%    2.536138e+05  2.337892e+07
50%    5.582480e+06  2.135789e+08
75%    5.098244e+07  3.488812e+08
max    6.941068e+10  8.868082e+10
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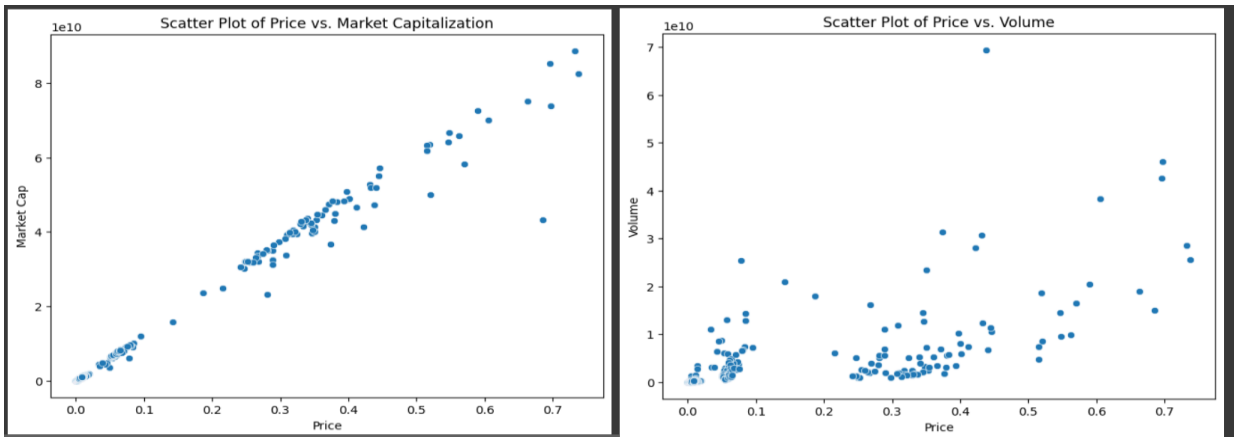
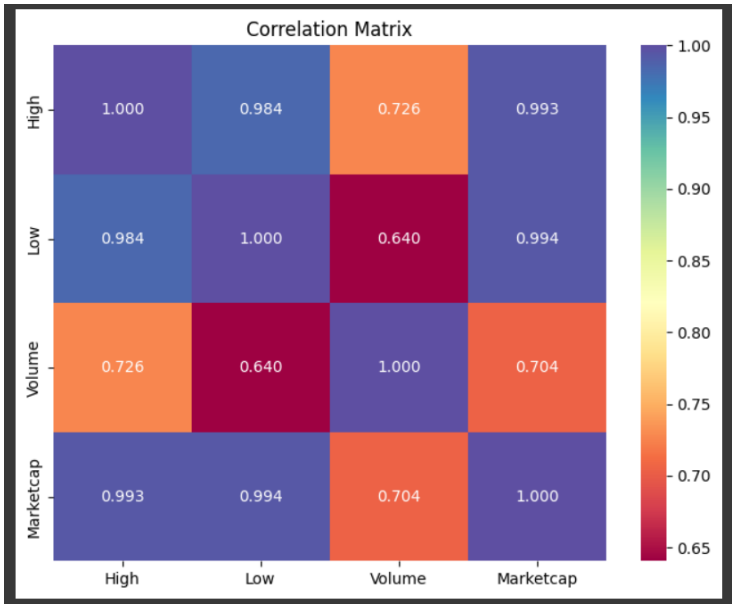
- Volume and Market Cap have a positive correlation between them, as the price increases market capitalization increases as well.
- Price and Volume does not seem to have any relationship between them unlike the left scatter plot with price and market cap.

- The datapoints in right plot are scattered across the plot with no pattern. This suggests that price does not necessarily changes in trading volume.



- The correlation coefficient between High and Low is 0.984 indicating very high correlation, this shows that High and Low prices of dogecoin is closely related.
- The correlation coefficient of High and Volume is 0.726 suggesting a moderate positive correlation, means when the price go high there is a tendency for trading volume to also increase.
- The correlation coefficient of High and Market Cap is 0.993 indicating a very positive correlation. As shown in early scatter plot.



Statistical Analysis: Statistical analysis such as T-tests, Analysis of Variance(ANOVA) has been done on the specific features of the dataset, key findings are as follows:

- One Sample Test: The One Sample test has been done for the 'High' column of the dataset with population mean of 0.14 which has given the results of t-statistic: 0.733, p-value: 0.463. The T test statistic measures the difference between sample mean and the

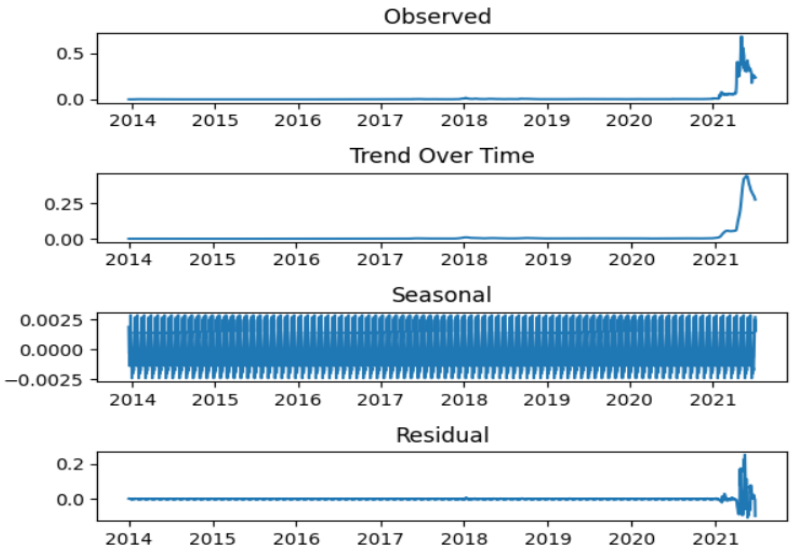
hypothesis mean. The value of 0.733 means that the sample mean in 0.733 standard errors away from the hypothesis mean. The p-value is 0.464 which represents the probability is much higher than the typical value i.e. 0.05, this suggests that there is not much of difference between the population mean and the sample mean.

- Two Sample Test: Two Sample test has been done for the 'High' and 'Low' column of the dataset. The results of t-statistic is: 1.51 with p-value being 0.129. The T test statistic suggest that the difference between the means of High and Low is 1.518 standard error away from 0 which shows that two groups are not significantly different . The p-value is 0.129 which represents the probability is much higher than the typical value i.e. 0.05, this suggests that there is not much of difference between the High and Low values.

- Paired T test: A paired T Test compare mean of two related group. A Paired T Test has been done on the 'Open' and 'Close' group. The results are as follows: T-statistic is –0.534 and the p-value is 0.593. The T-statistic value suggests that the difference between mean of Open and Close is very small. However, the negative sign shows that the value of Close is higher than the value of Open. The p-value being 0.534 is way more than the suggest value of 0.05 which suggest that the two columns are highly related to each others.

- **Analysis of Variance(ANOVA):** The ANOVA analysis is being done on three or more features to see if the mean for one group is significantly different from the other two. The analysis has been done on High, Low and Open price of the crypto. The F-Statistic value is 0.77 and the P-Value is 0.509. A higher F statistic value typically indicates that there are significant differences between the group means but in this case the value is 0.77 which is relatively low. This low value suggest that the variance between the groups are not large enough, means there is not a significant different between these three features. A Higher P value i.e. value greater than 0.05 also suggest that there are not enough evidence to say that there are differences between means of High, Low and Open prices of the cryptocurrency in the dataset.

Time Series Analysis: Time-series decomposition is a technique used to break down a time series into its constituent components such as trends, seasonality, and residuals. This helps in understanding the underlying patterns in the data. A time series decomposition has been applied to the



- **Methodology**: Additive decomposition was used, assuming that the components add up to the observed data. A period of 30 days was chosen to capture monthly seasonality.
- **Results**: The "Trend Over Time" component shows a general up and down movement in Dogecoin prices over time. The seasonal component reveals recurring patterns, such as periodic price fluctuations. The residuals represent noise or irregular fluctuations that cannot be explained by trend or seasonality.

Conclusion: The EDA reveals a strong positive relationship between price and market capitalization. The correlation matrix quantifies these relationships, showing strong correlations between price and market cap and moderate correlations between price, volume, and market capitalization. While analyzing the Univariate most of the variables are positively skewed with values grouping near low ranges and occasionally spikes and peaks. the price of the cryptocurrency, volume and market cap shows high variability in the later period of time.

Time series analysis shows a comprehensive workflow for analyzing and forecasting Dogecoin prices using time series techniques. Major steps including preprocessing the data, decomposing it into trend, seasonality, and residuals, and evaluating the performance of ARIMA and moving average models. While ARIMA provided better accuracy, the moving average model offered simplicity and ease of interpretation. The ANOVA test indicates no significant differences between the High, Low, and Open prices, while the t-test helped assessing differences between two specific groups. Combining these tests provide valuable insights into the relationships and significance of variables in the dataset.